

First Distributed PDL Monitoring on Coherent DSP Chip in Pluggable Transceiver

Minami Takahashi ⁽¹⁾, Takeo Sasai ⁽¹⁾, Etsushi Yamazaki ⁽¹⁾

⁽¹⁾ Network Innovation Laboratories, NTT, Inc., minami.takahashi@ntt.com

Abstract We demonstrate the first experimental localization and quantification of distributed PDL, fully implemented on a coherent DSP in a 118-GBd OSFP transceiver. Over a WDM link, on-chip digital-SOP-optimized polarization-resolved power profiles identify multiple PDL elements solely from ZR+ traffic signals, with <1.0-dB estimation error. ©2026 The Author(s)

Introduction

As coherent pluggable transceivers extend their reach from data-centre interconnect to metro and long-haul networks [1], embedded monitoring functions become increasingly important for efficient network operation. Current coherent pluggable modules report cumulative link metrics such as chromatic dispersion, differential group delay, and polarization-dependent loss (PDL), as defined in C-CMIS [2]. However, these metrics lack spatial resolution and cannot identify the locations of performance degradations along multi-span links.

Longitudinal power monitoring (LPM) addresses this limitation by reconstructing distance-resolved power profiles solely from received signals at coherent receivers, without dedicated instruments such as OTDRs [3,4]. Recently, LPM was implemented on a coherent DSP chip inside an OSFP module [5], demonstrating that distance-resolved monitoring can be realized directly within the transceiver. However, this on-chip implementation estimated only the total power profile and did not provide polarization-resolved information.

PDL, introduced by in-line optical components such as wavelength-selective switches and amplifiers, is a dominant impairment limiting the capacity of polarization-multiplexed systems [6,7]. Due to the absence of methods for measuring distributed PDL during operation, network designs assume a worst-case PDL margin of 4–6 dB [8]. If the locations and magnitudes of excessive PDL sources could be identified on a per-link basis, this margin could be reduced based on actual PDL measurements rather than worst-case assumptions. Several studies have demonstrated distributed PDL estimation using LPM with offline DSP [9–12]. However, experimental validation has been limited to a single PDL element [9–11], and localization and quantification of multiple PDL elements has only been demonstrated in simulation [11,12]. Moreover, none of these have been implemented on a coherent DSP chip.

In this paper, we demonstrate the first experimental localization and quantification of multiple

PDL elements, fully implemented on a coherent DSP chip inside a pluggable transceiver. The on-chip DSP implementation combines polarization-resolved power estimation with fully digital SOP sweeping, enabling identification of distributed PDL solely from standard ZR+ traffic signals. Experiments over a 200-km WDM link at the system optimum power verify localization and quantification of multiple PDL elements with estimation error below 1.0 dB, while single-PDL estimation achieves errors below 0.5 dB.

Principle and On-Chip Implementation

LPM estimates a distance-resolved power profile from the received signal by exploiting the signal-power dependence of the Kerr effect [3,4]. In standard LPM, a single profile is obtained from the total power across both polarizations. In this work, the on-chip LPM engine in [5] is extended to polarization-resolved profiles $\gamma'_p(z) \propto P_p(z)$ for each polarization $p \in \{x, y\}$, where P_p denotes the per-polarization signal power.

In conventional LPM [4], the least-squares estimate of the profile is generally formulated as

$$\hat{\gamma}' = (\text{Re}[G^\dagger G])^{-1} \text{Re}[G^\dagger A_1] \quad (1)$$

where G is a matrix describing the nonlinear perturbation at each spatial position and A_1 is the first-order perturbation of the received signal. This estimation is applied to each polarization independently to obtain $\gamma'_p(z)$. Although this algorithm can estimate polarization-wise power profiles, it does not mean that the correct PDL can be estimated. This is because the power loss in each polarization depends on the incident SOP. The engine therefore performs a digital SOP sweep within the receiver by jointly rotating the received and reference signals, scanning for the signal basis that maximizes the polarization power difference. When multiple PDL elements with different PSP orientations are present, this process is repeated for each element [12]. In this way, PDL values can be estimated entirely in the DSP without optical SOP manipulation.

However, computing (1) requires storing the matrix G , whose memory footprint ranges from megabytes to gigabytes depending on the link length and spatial resolution [5]. For PDL

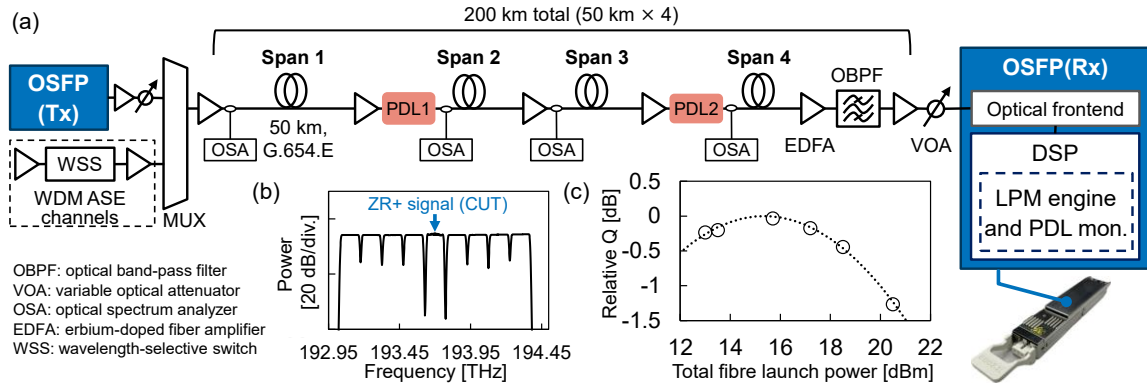


Fig. 1: (a) Experimental setup. A 400ZR+ QPSK signal at 118 GBd is transmitted over a 200-km link consisting of four 50-km spans of G.654.E fibre. PDL emulators are inserted at the inputs of spans 2 and 4 to emulate a link with distributed PDL. Inset: OSFP transceiver used as both Tx and Rx. (b) WDM optical spectrum. The channel under test (CUT) is at 193.7 THz. (c) Relative Q-factor vs. total fibre launch power. All experiments are conducted at the optimum power of 15.7 dBm.

monitoring, this computation must be performed independently for each polarization and repeated across multiple SOPs, further multiplying the memory and computational requirements. A direct implementation of (1) on a DSP is therefore prohibitive. To address this, the on-chip LPM engine employs an analytical-form-assisted approach inspired by [13], in which the analytical form eliminates the need to store G , reducing the memory requirement by orders of magnitude. This enables the LPM-based PDL monitoring to be implemented entirely on the DSP chip, and no measurable increase in module power consumption is observed.

Experimental Setup

Fig. 1(a) shows the experimental setup. The transmitter is an OSFP transceiver generating a 400ZR+ signal (QPSK, 118 GBd). The signal is combined with eight ASE-emulated WDM channels at 150-GHz spacing, giving a total of nine channels, as shown in Fig. 1(b). The transmission link consists of four 50-km spans of G.654.E fibre with an attenuation coefficient of 0.16 dB/km and an effective area of $110 \mu\text{m}^2$ at 1550 nm. EDFAs provide in-line amplification, and their gains are adjusted to maintain a constant fibre launch

power at each span input regardless of the insertion loss and PDL setting of the emulators. Two PDL emulators are inserted at the inputs of spans 2 and 4 to emulate a link with distributed PDL. The signal is received by an OSFP of the same type, and the polarization-resolved LPM is executed on the coherent DSP chip within the receiver module with a spatial step size of 1.7 km. The fibre launch power is set to the optimum of 15.7 dBm, as shown in Fig. 1(c). Post-FEC error-free operation is confirmed during all LPM executions. No dedicated probe or training sequence is used, and no manual polarization adjustment is required; the SOP optimization is performed entirely in the digital domain within the Rx DSP chip, even in the presence of SOP variations.

Results

Fig. 2(a) shows the polarization-resolved power profiles obtained on-chip when a single PDL of 3 dB (PDL1) is inserted at the input of span 2. The SOP sweep angle is optimized to maximize the power difference between the two polarizations. A clear separation between the x- and y-polarization profiles is observed starting at the PDL insertion point, whereas the profiles overlap before that point, confirming successful on-chip

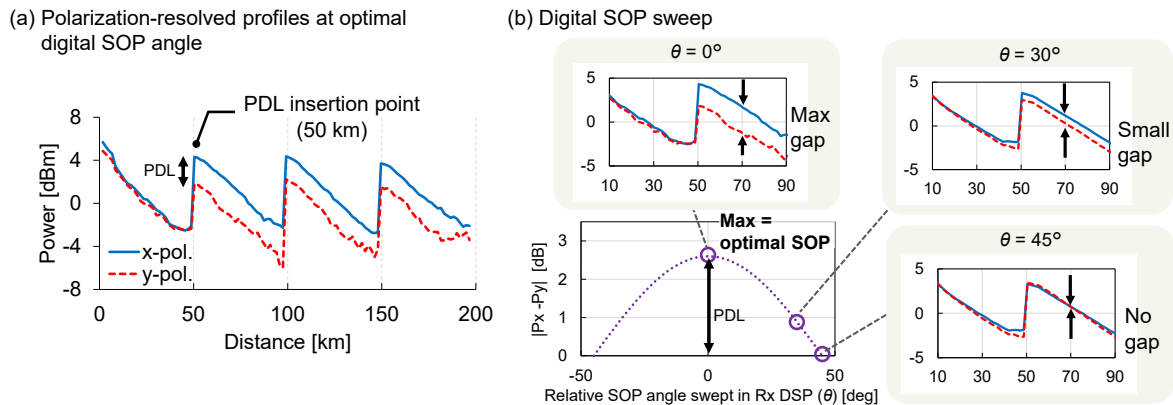


Fig. 2: (a) Polarization-resolved power profiles obtained on-chip with a single PDL of 3 dB at the input of span 2. The SOP sweep angle is optimized to maximize the polarization power difference. (b) Power difference between x- and y-polarization profiles as a function of digital SOP sweep angle. The curve is a fitted approximation. Insets show experimentally obtained profiles near the PDL insertion point (10–90 km) at selected angles.

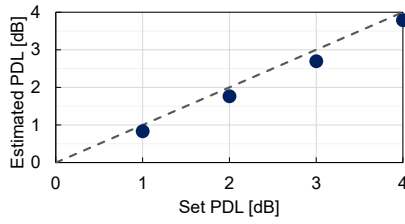


Fig. 3: Estimated vs. set PDL values for a single PDL element at the span-2 input (1–4 dB). The dashed line represents ideal estimation.

localization of the PDL element. Fig. 2(b) shows the power difference between polarizations as a function of the SOP sweep angle. The power difference varies with the sweep angle and reaches its maximum when the signal basis aligns with the PSP of the PDL element. By selecting the angle that maximizes the polarization power difference, the PDL value can be estimated on chip.

To evaluate the estimation accuracy for a single PDL element, the PDL value (PDL1) was varied from 1 to 4 dB at the input of span 2 while span 4 had no PDL. Fig. 3 shows the estimated PDL, obtained by averaging the polarization power difference over the affected span, as a function of the set value. The estimated values show good agreement with the set values, with estimation errors below 0.5 dB across all settings. These results indicate that the on-chip monitoring provides reliable quantification of a single PDL element under the present experimental conditions, including SOP variations.

Finally, we demonstrate localization and quantification of two PDL elements, both set to 3 dB. Because the PSP orientations of the two PDL elements generally differ owing to polarization rotation in the fibre, the SOP sweep is optimized separately for each element [12]. Fig. 4(a) shows the polarization-resolved profiles over 0–150 km

when the SOP is optimized for the first PDL element; a clear power difference is observed from span 2 onward. After correcting for the PDL1-induced offset, Fig. 4(b) shows the profiles over 50–200 km with the SOP optimized for the second element, revealing a clear power difference at span 4. Figs. 4(c) and 4(d) show the corresponding PDL indicators. In both cases, a clear step is observed at the respective PDL insertion point, with an estimated PDL of approximately 2.3 dB for a set value of 3 dB. Although the quantitative accuracy is lower than in the single-PDL case, partly due to the limited number of SOP sweep angles and profile averaging in the on-chip implementation, the results confirm that both PDL elements can be separately localized and detected on chip.

Conclusion

We demonstrated the first on-chip localization and quantification of distributed PDL in a coherent 118-GBd standard-compliant OSFP transceiver. By combining polarization-resolved estimation with digital SOP sweeping entirely within the receiver DSP, multiple PDL elements were localized and quantified over a WDM link solely from ZR+ traffic signals, which is, to the best of our knowledge, the first such experimental demonstration using LPM. For a single PDL element, estimation errors below 0.5 dB were achieved, while for multiple elements, SOP optimization for each element enabled their separate localization and detection despite different PSP orientations. These results indicate that distributed PDL monitoring can be realized as a native function of coherent pluggable transceivers and help reduce PDL margins in optical network design.

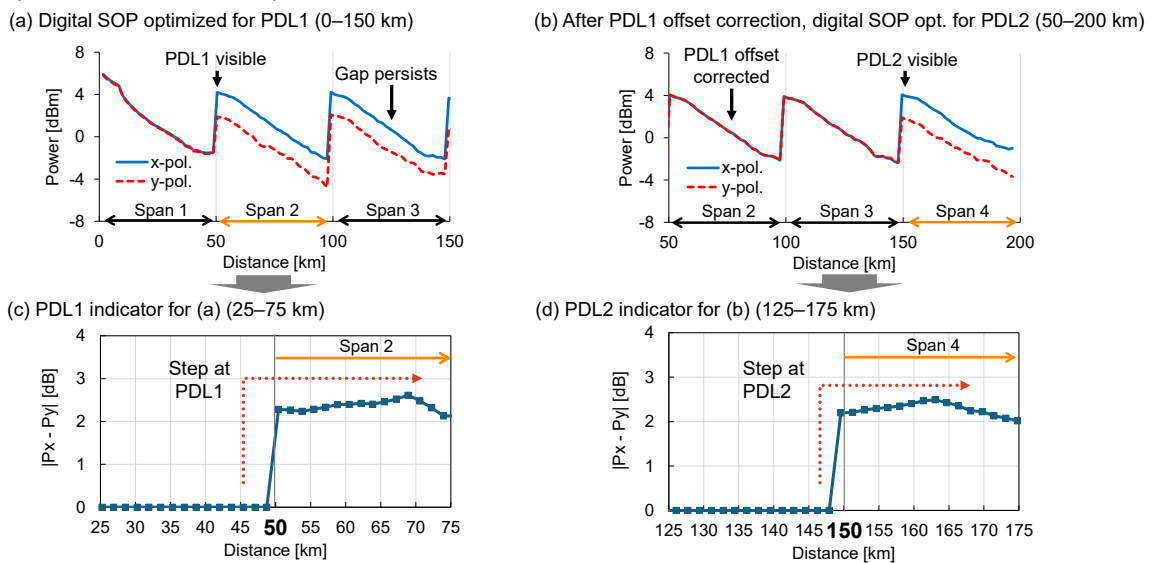


Fig. 4: Localization and quantification of two PDL elements (PDL1 at the span-2 input and PDL2 at the span-4 input, both set to 3 dB). (a) Polarization-resolved profiles with the SOP optimized for PDL1, showing a clear power difference from span 2 onward. (b) Profiles after PDL1 offset correction with the SOP optimized for PDL2, revealing a clear power difference at span 4. (c), (d) Corresponding PDL indicators near the PDL insertion point, defined as the difference between the x- and y-polarization profiles.

Acknowledgements

These results were obtained from the commissioned research (No. JPJ012368G60201) by National Institute of Information and Communications Technology (NICT), Japan. We thank the members of NTT Innovative Devices Corporation for their support.

References

- [1] J. Pedro, M. M. Hosseini, and A. Napoli, "Extended network applications of coherent pluggable transceivers," *Journal of Optical Communications and Networking*, vol. 17, no. 2, pp. A210–A223, 2025, DOI: 10.1364/JOCN.538690.
- [2] OIF-C-CMIS-01.4, "Implementation Agreement for Coherent CMIS," Optical Internetworking Forum, April 2025.
- [3] T. Tanimura, S. Yoshida, K. Tajima, S. Oda, and T. Hoshida, "Fiber-longitudinal anomaly position identification over multi-span transmission link out of receiver-end signals," *Journal of Lightwave Technology*, vol. 38, no. 9, pp. 2726–2733, 2020, DOI: 10.1109/JLT.2020.2984270.
- [4] T. Sasai, M. Takahashi, M. Nakamura, E. Yamazaki, and Y. Kisaka, "Linear least squares estimation of fiber-longitudinal optical power profile," *Journal of Lightwave Technology*, vol. 42, no. 6, pp. 1955–1965, 2024, DOI: 10.1109/JLT.2023.3339053.
- [5] T. Sasai, Y. Nakajima, M. Takahashi, R. Kaneko, M. Nakamura, A. Matsushita, F. Hamaoka and E. Yamazaki, "First coherent DSP and pluggable transceiver capable of distance-resolved, longitudinal power monitoring," in *Optical Fiber Communication Conference (OFC) 2026*, San Francisco, USA, 2026.
- [6] M. Shtaiif, "Performance degradation in coherent polarization multiplexed systems as a result of polarization dependent loss," *Optics Express*, vol. 16, no. 18, pp. 13918–13932, 2008, DOI: 10.1364/OE.16.013918.
- [7] A. El Amari, N. Gisin, B. Perny, H. Zbinden, and C. W. Zimmer, "Statistical prediction and experimental verification of concatenations of fiber optic components with polarization dependent loss," *Journal of Lightwave Technology*, vol. 16, no. 3, pp. 332–339, 1998, DOI: 10.1109/50.661372.
- [8] OpenROADM MSA, "OpenROADM device white paper, optical specifications," v7.0.
- [9] M. Eto, K. Tajima, S. Yoshida, S. Oda, and T. Hoshida, "Location-resolved PDL monitoring with Rx-side digital signal processing in multi-span optical transmission system," in *Optical Fiber Communication Conference (OFC) 2022*, San Diego, USA, 2022, paper Th1C.2, DOI: 10.1364/OFC.2022.Th1C.2.
- [10] A. May, E. Awwad, P. Ramantanis, and P. Ciblat, "Receiver-based localization and estimation of polarization dependent loss," in *27th OptoElectronics and Communications Conference (OECC 2022)*, Toyama, Japan, 2022, DOI: 10.23919/OECC/PSC53152.2022.9849907.
- [11] L. Andrenacci, G. Bosco, and D. Piliro, "PDL localization and estimation through linear least squares-based longitudinal power monitoring," *IEEE Photonics Technology Letters*, vol. 35, no. 24, pp. 1431–1434, 2023, DOI: 10.1109/LPT.2023.3331110.
- [12] M. Takahashi, T. Sasai, E. Yamazaki, and Y. Kisaka, "Monitoring PDL value and location using DSP-based longitudinal power profile estimation," *Journal of Lightwave Technology*, vol. 42, no. 17, pp. 5866–5873, 2024, DOI: 10.1109/JLT.2024.3407722.
- [13] T. Sasai and E. Yamazaki, "Closed-form expression for spatial correlation function of nonlinear fiber propagation and its applications to longitudinal power monitoring," in *OFC 2026*, San Francisco, USA, 2026, paper Tu3A.5.